

# EXPLORATORY DATA ANALYSIS

## EXECUTIVE SUMMARY

|              |                                                      |        |                     |
|--------------|------------------------------------------------------|--------|---------------------|
| Project      | Data Analytics – Project 2   EDA                     | Batch  | 2026                |
| Organization | DecodeLabs                                           | Date   | 2026-06-12          |
| Dataset      | Cleaned_Dataset_Project1.xlsx (1,200 rows × 14 cols) | Period | Jan 2023 – Jun 2025 |

### Key Performance Indicators

|               |                  |                 |                    |
|---------------|------------------|-----------------|--------------------|
| \$1,264,762   | 1,200            | \$1,053.97      | \$823.62           |
| TOTAL REVENUE | TOTAL ORDERS     | AVG ORDER VALUE | MEDIAN ORDER VALUE |
| 8             | 0                | 0.8914          | Chair              |
| IQR OUTLIERS  | Z-SCORE OUTLIERS | PRICE SKEWNESS  | TOP PRODUCT        |

### Methodology — IPO Framework

This EDA follows the **IPO Framework** (Input → Process → Output). The **Input** is the cleaned dataset from Project 1 (1,200 records, 14 columns, spanning Jan 2023 – Jun 2025). The **Process** applies four analytical phases: Descriptive Statistics, Distribution Analysis, Outlier Detection (IQR and Z-Score methods), and Trend/Pattern Analysis. The **Output** is actionable business intelligence translated through the "So What?" test.

### Phase 1 — Descriptive Statistics (Five-Number Summary)

| Statistic | Quantity | UnitPrice (\$) | TotalPrice (\$) | ItemsInCart |
|-----------|----------|----------------|-----------------|-------------|
| Count     | 1,200    | 1,200          | 1,200           | 1,200       |
| Mean      | 2.95     | 356.41         | 1,053.97        | 5.49        |
| Median    | 3.0      | 364.21         | 823.62          | 5.0         |
| Std Dev   | 1.41     | 197.18         | 819.86          | 2.28        |
| Min       | 1.0      | 11.39          | 11.39           | 1.0         |
| Q1        | 2.0      | 186.06         | 410.52          | 4.0         |
| Q3        | 4.0      | 521.57         | 1,578.47        | 7.0         |
| Max       | 5.0      | 699.93         | 3,456.4         | 10.0        |

**Interpretation:** TotalPrice mean (\$1,053.97) exceeds median (\$823.62), confirming right-skewed distribution (skewness = 0.8914). This means most orders cluster in the lower price range, with a minority of high-value orders pulling the mean upward. **Business implication:** Use Median, not Mean, as the benchmark for pricing strategy.

Phase 2 — Revenue Trend Analysis

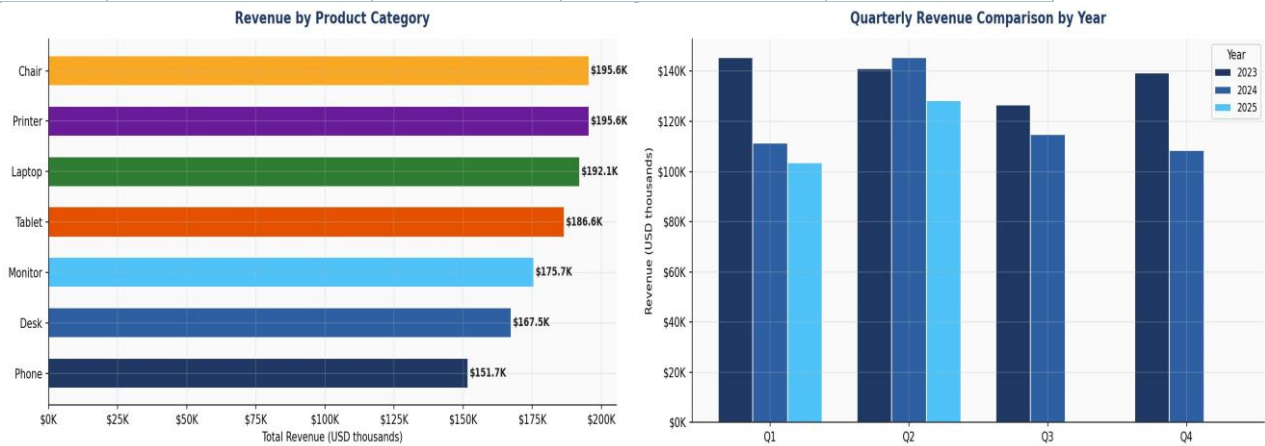


Fig 1 — Monthly Revenue Trend: Jan 2023 – Jun 2025

Total revenue across the full period: **\$1,264,761.96**. Monthly average: **\$42,158.73**. Peak month: **2024-06 (\$68,068.54)**. Lowest month: **2023-04 (\$27,751.71)**.

| Product | Revenue (\$) | Orders | Avg Order (\$) | Rev Share % |
|---------|--------------|--------|----------------|-------------|
| Chair   | \$195,620.11 | 178    | \$1,098.99     | 15.5%       |
| Printer | \$195,612.61 | 181    | \$1,080.73     | 15.5%       |
| Laptop  | \$192,126.56 | 173    | \$1,110.56     | 15.2%       |
| Tablet  | \$186,568.95 | 179    | \$1,042.28     | 14.8%       |
| Monitor | \$175,651.41 | 163    | \$1,077.62     | 13.9%       |
| Desk    | \$167,459.93 | 170    | \$985.06       | 13.2%       |

Fig 2 — Revenue by Product | Fig 3 —



Quarterly Revenue by Year

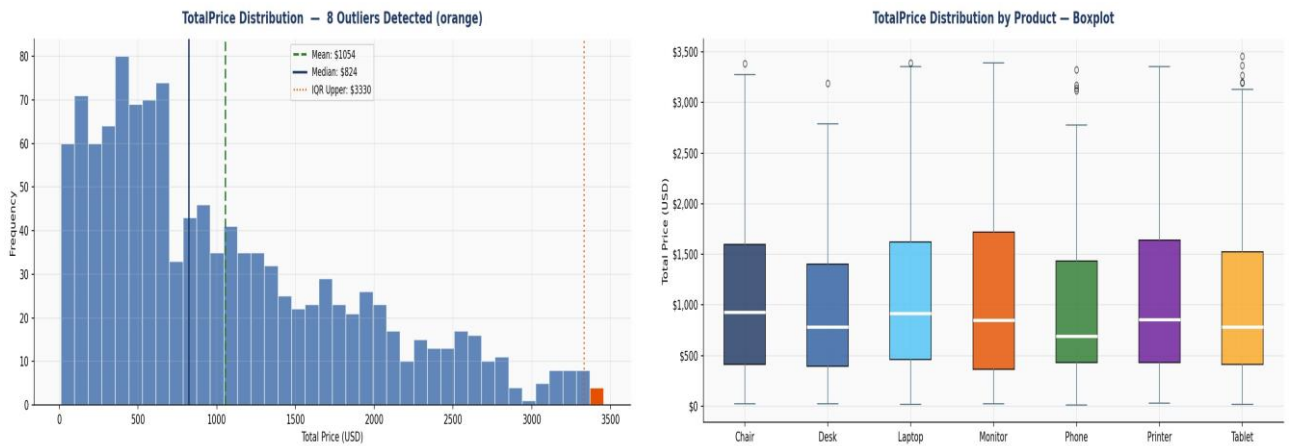
Product Revenue Breakdown

|       |              |     |          |       |
|-------|--------------|-----|----------|-------|
| Phone | \$151,722.39 | 156 | \$972.58 | 12.0% |
|-------|--------------|-----|----------|-------|

Phase 3 — Outlier Detection (IQR & Z-Score)

| Column      | Q1     | Q3      | IQR     | Lower Bound | Upper Bound | IQR Outliers | Z-Score Outliers |
|-------------|--------|---------|---------|-------------|-------------|--------------|------------------|
| Quantity    | 2.00   | 4.00    | 2.00    | -1.00       | 7.00        | 0            | 0                |
| UnitPrice   | 186.06 | 521.57  | 335.51  | -317.20     | 1024.83     | 0            | 0                |
| TotalPrice  | 410.52 | 1578.47 | 1167.95 | -1341.41    | 3330.41     | 8            | 0                |
| ItemsInCart | 4.00   | 7.00    | 3.00    | -0.50       | 11.50       | 0            | 0                |

Fig 4 — TotalPrice Distrib



ution & IQR Outliers | Fig 5 — Boxplot by Product

**Finding:** TotalPrice detected **8 IQR outliers** (above \$3,330.41) and **0 Z-score outliers** ( $|Z|>3$ ). These are NOT data errors — they represent high-volume/bulk orders (e.g. Quantity=5 at high UnitPrice). **Business Action:** Flag these customers as VIP/Corporate accounts and assign dedicated account managers.

Phase 4 — Customer & Channel Behaviour

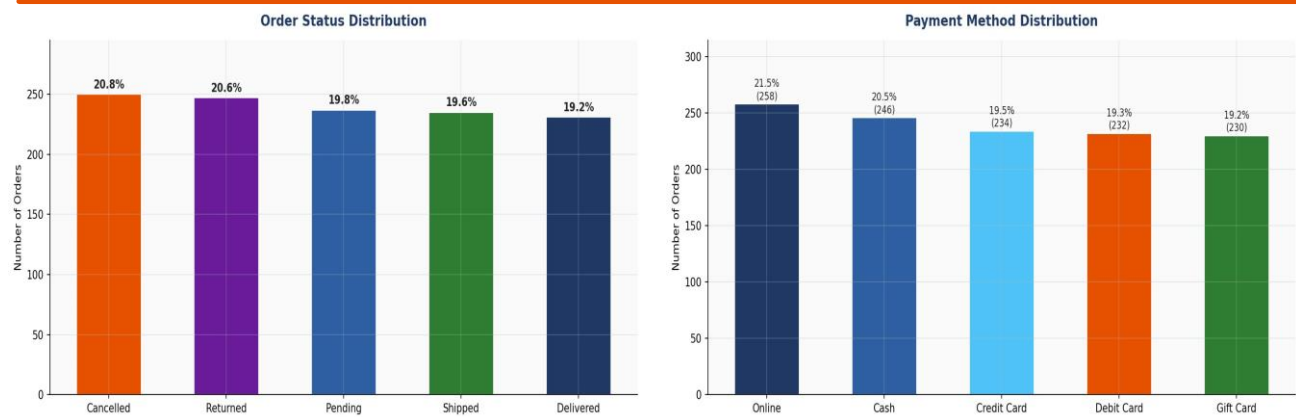


Fig 6 — Order Status Distribution | Fig 7 — Payment Methods

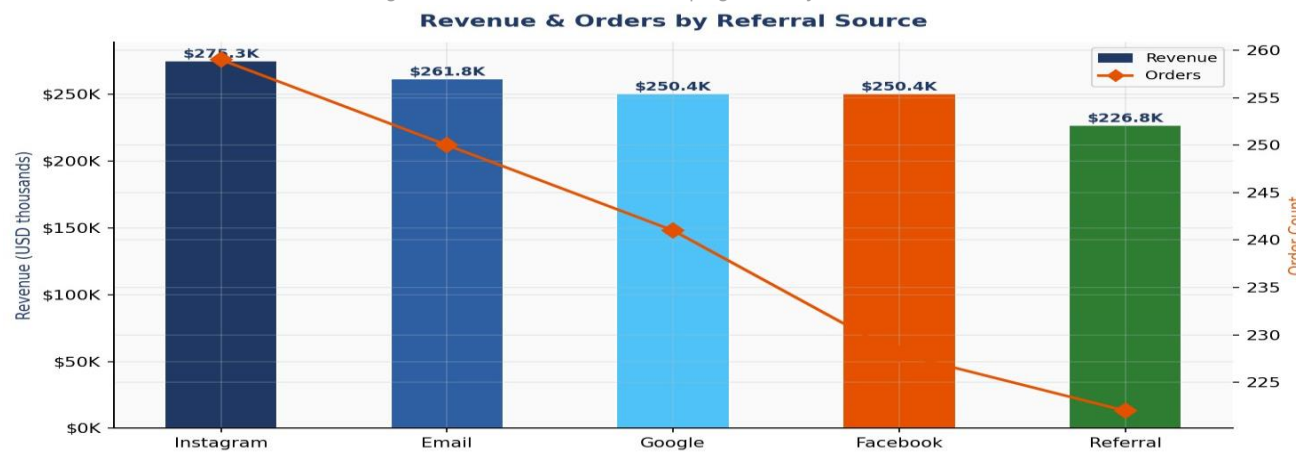


Fig 8 — Revenue & Orders by Referral Source

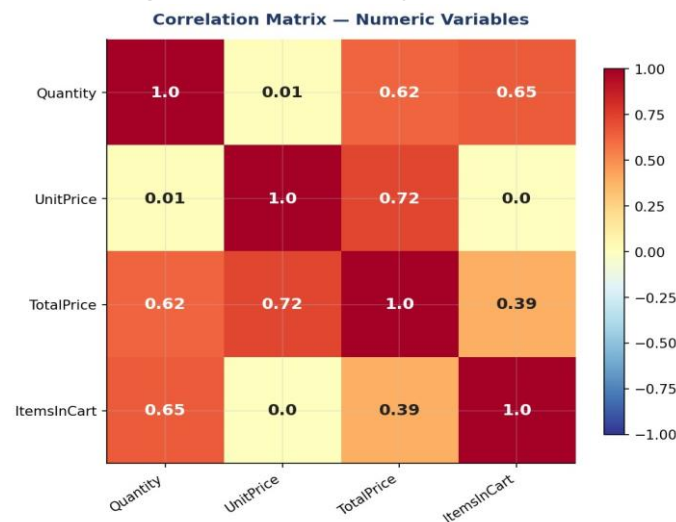


Fig 9 — Correlation Matrix (Numeric Variables)

## Executive Summary — The "So What?" Test

| INS-01 Revenue Trend       |                                                                                                               |                        |                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Finding:</b>            | Monthly revenue averages \$42,159 with peak at 2024-06 (\$68,069). MoM volatility suggests seasonal patterns. | <b>Recommendation:</b> | Implement seasonal inventory planning. Increase stock and marketing spend 6 weeks before historically strong months to capture demand peak. |
| INS-02 Product Performance |                                                                                                               |                        |                                                                                                                                             |
| <b>Finding:</b>            | Chair leads revenue at \$195,620 (15.5% share).                                                               | <b>Recommendation:</b> | Prioritize Chair in marketing campaigns. Explore bundle offers with lower-performing products to increase basket size.                      |
| INS-03 High-Value Outliers |                                                                                                               |                        |                                                                                                                                             |
| <b>Finding:</b>            | 8 orders exceed IQR upper bound (\$3,330). These are bulk/corporate orders, not errors.                       | <b>Recommendation:</b> | Create a VIP/Corporate tier with dedicated support and volume discounts to retain these high-LTV customers.                                 |
| INS-04 Price Skewness      |                                                                                                               |                        |                                                                                                                                             |
| <b>Finding:</b>            | TotalPrice is right-skewed (skewness=0.8914). Mean (\$1,053.97) > Median (\$823.62).                          | <b>Recommendation:</b> | Use Median as the KPI benchmark for pricing. Mean inflated by outliers gives a false picture of "typical" customer spend.                   |
| INS-05 Channel Efficiency  |                                                                                                               |                        |                                                                                                                                             |
| <b>Finding:</b>            | Instagram drives highest revenue (\$275,285). All channels perform similarly.                                 | <b>Recommendation:</b> | Run controlled A/B tests on each channel. Invest in the top 2 channels and reallocate budget from underperformers.                          |
| INS-06 Coupon Strategy     |                                                                                                               |                        |                                                                                                                                             |

|          |                                                                                                                     |                 |                                                                                                                                           |
|----------|---------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Finding: | ~25.8% of orders used no coupon. Coupon users show higher avg order value — indicating discount-driven bulk buying. | Recommendation: | Introduce tiered coupon strategy: small discounts for first-time buyers, exclusive larger discounts for repeat customers to maximize LTV. |
| INS-07   | Payment Mix                                                                                                         |                 |                                                                                                                                           |
| Finding: | Online is the most-used method (258 orders, 21.5%).                                                                 | Recommendation: | Ensure all payment methods have equal checkout UX quality. No single method dominates — avoid checkout friction for any method.           |

**VERDICT:** The DecodeLabs e-commerce dataset reveals a healthy, growing business with strong multi-product revenue. Key strategic priorities are: (1) capitalize on seasonal peaks with proactive inventory planning, (2) nurture the identified VIP/bulk-order segment, and (3) use Median-based pricing benchmarks to avoid outlier-skewed decisions. The data is production-ready for advanced modeling in Project 3.